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Getting to Know Your Data

"Truth, like gold, is to be obtained not by its growth, but by washing away from it all that is not gold."

—Leo Tolstoy

In this chapter, we explore features within the Databricks DI Platform that help improve and monitor data quality and facilitate data exploration. There are numerous approaches to getting to know your data better with Databricks. First, we cover how to oversee data quality with **Delta Live Tables** (**DLT**) to catch quality issues early and prevent the contamination of entire pipelines. We'll take our first close look at Lakehouse Monitoring, which helps us analyze data changes over time and can alert us to changes that concern us. Lakehouse Monitoring is a big time-saver, allowing you to focus on mitigating or responding to data changes rather than creating notebooks that calculate standard metrics.

Moving on to data exploration, we will look at a couple of low-code approaches: Databricks Assistant and **AutoML**. Finally, we will touch on embeddings. We created embeddings from chunks of text in *Chapter 3*, and you'll learn how to use Databricks **Vector Search** (**VS**) to explore your embeddings.

Here is what you will learn as part of this chapter:

- Improving data integrity with DLT
- Monitoring data quality with Databricks Lakehouse Monitoring
- Exploring data with Databricks Assistant
- Generating data profiles with AutoML
- Preparing data for vector search and database indexing
- Enhancing data retrieval with Databricks Vector Search
- Applying our learning